Summary

The digestive system of human consists of the mouth (oral cavity), pharynx, esophagus, stomach, small intestine, large intestine, rectum and anus. The accessory digestive glands include the salivary glands, the liver (with gall bladder) and the pancreas. The process of digestion involves intake of the food (Ingestion), breakdown of the food into micromolecules (Digestion), absorption of these molecules into the blood stream (Absorption), the absorbed substances becoming components of cells (Assimilation) and elimination of the undigested substances (Egestion).

Food comprises of macronutrients and micronutrients. The nutrients required in larger quantities are called macronutrients, whereas those required in small quantities are called micronutrients. Essential nutrients cannot be synthesized by the body; they have to be included in the diet. Macronutrients are lipids, carbohydrates, proteins and the micronutrients are vitamins and minerals. Water plays an important role in the metabolic processes and prevents dehydration of the body.

Intestinal tract is more prone to bacterial, viral and parasitic worm infections. Such infection, called colitis, may couse inflammation of the inner lining of the colon. Growing children require more protein for their growth and development. Protein deficient diet during early stage of children may lead to protein energy malnutrition such as Marasmus and Kwashiorkor.



Alimentary canal faces a conflict between the need of nutrient absorption and to keep our intestinal tract free from pathogenic bacteria and virus. About 7 litres of digestive juice are poured into the alimentary canal and are reabsorbed each day. If this does not happen the body gets rapidly dehydrated and may lead to reduction in the blood pressure.

Activity

Test for Starch: Add a few drops of iodine to the given warm food sample. If any starch is present in the given food sample it will change the colour of the iodine from brown to blue-black.

Test for protein: Mix the given food sample with 3mL of water in a test tube. Shake the mixture, and then add a few drops of Biuret solution. If protein is present, the colour of the solution will change to purple.

Test for glucose: Mix the given food sample with 3mL of water in a test tubes. Shake the mixture, and then add a few drops of Benedict's solution. Keep the test tube in a water bath and heat carefully. If glucose is present, the colour of the solution will change from blue to green to brick red depending upon the amount of glucose.

XI Std Zoology Chapter-5 EM.indd 89 30/12/2022 10:38:49